MP363 – Quantum Mechanics I

Problem Set 4

Due by the end of lecture on Friday, 4 December 2015

1. The energy eigenfunctions for the infinite square well are $\phi_n(t,x) = u_n(x)e^{-iE_n t/\hbar}$ for $n = 1, 2, \ldots$, where

$$E_n = \frac{\pi^2\hbar^2}{2ma^2}n^2, \qquad u_n(x) = \begin{cases} \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi x}{a}\right) & 0 \le x \le a, \\ 0 & x < 0, x > a \end{cases}$$

Suppose the state of a particle in this well is given by the wavefunction

$$\psi(t,x) = \frac{1}{\sqrt{5}}\phi_1(t,x) - \frac{2i}{\sqrt{5}}\phi_3(t,x).$$

(a) Remembering that the inner product between two functions χ and ζ for the infinite square well is given by

$$\langle \chi \mid \zeta \rangle = \int_0^a \chi^*(t,x)\zeta(t,x)\,\mathrm{d}x,$$

verify that the state ψ has unit norm.

(b) If the energy of this state is measured, what are the possible results, and with what probabilities do they occur?

(c) Find the probability, as a function of time, that the particle is found in the right half of the well, i.e. between $x = a/2$ and $x = a$.

2. Suppose a system is in the N^{th} eigenstate of an infinite square well, namely, the wavefunction $\psi(t,x)$ is zero for $x \le 0$ and $x \ge a$, and

$$\psi(t,x) = \sqrt{\frac{2}{a}}e^{-iE_N t/\hbar}\sin\left(\frac{N\pi x}{a}\right), \qquad E_N = \frac{\pi^2\hbar^2}{2ma^2}N^2$$

for $0 < x < a$.

(a) The position and momentum operators act on an arbitrary function $\zeta(t,x)$ via

$$X\zeta = x\zeta, \qquad P\zeta = -i\hbar\frac{\partial\zeta}{\partial x}.$$

Use these to show that, for the given state ψ, the expectation value of X is $a/2$ and the expectation value of P is zero.

(b) Find $\langle X^2 \rangle$ and $\langle P^2 \rangle$.

(c) The uncertainty in the measurement of an observable A is

$$\Delta a = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}.$$

Using this and your results from (a) and (b), find expressions for Δx and Δp and use them to confirm that $\Delta x \Delta p$ is greater than $\hbar/2$ for all values of N.

3. Suppose the wavefunction for a particle in an infinite square well is given (for $0 < x < a$) by

$$\psi(x) = Cx(a - x)$$

where C is some constant.

(a) Determine the value of C such that ψ is normalised.

(b) Use the actions of X and P as given above, together with the action of the Hamiltonian

$$H\zeta = -\frac{\hbar^2}{2m}\frac{\partial^2 \zeta}{\partial x^2},$$

to find the expectation values $\langle X \rangle$, $\langle X^2 \rangle$, $\langle P \rangle$, $\langle P^2 \rangle$ and $\langle H \rangle$ for the state ψ.

(c) Use your results from (b) to confirm that $\Delta x \Delta p$ is larger than $\hbar/2$ for this state.

(d) Find the probability that a measurement of energy gives $E_3 = 9\pi^2\hbar^2/2ma^2$.